

Matrix Implementation of Graph (Adjacency Matrix)

A graph can be represented using a **matrix** where:

- Rows represent vertices (nodes).
- Columns represent vertices (nodes).
- If an edge exists between two vertices, store **1**.
- Otherwise, store **0**.

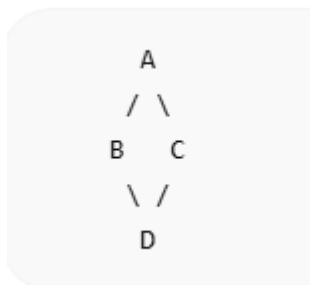
1. Initial Graph

Consider vertices:

A, B, C, D

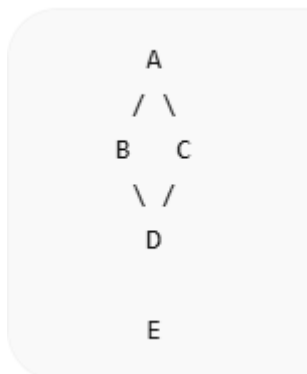
Edges:

- $A \rightarrow B$
- $A \rightarrow C$
- $B \rightarrow D$
- $C \rightarrow D$



	A	B	C	D
A	0	1	1	0
B	0	0	0	1
C	0	0	0	1
D	0	0	0	0

2. Insertion of a Node



New Matrix

Add one row and one column for E.

	A	B	C	D	E
A	0	1	1	0	0
B	0	0	0	1	0
C	0	0	0	1	0
D	0	0	0	0	0
E	0	0	0	0	0

3. Insertion of an Edge

Insert edge:

D → E



Change cell (D,E) from 0 to 1.

	A	B	C	D	E
A	0	1	1	0	0
B	0	0	0	1	0
C	0	0	0	1	0
D	0	0	0	0	1
E	0	0	0	0	0

4. Deletion of an Edge

Delete edge:

A → C

A
|
B
|
D
|
E

C
\
D

Matrix

Set cell (A,C) = 0

	A	B	C	D	E
A	0	1	0	0	0
B	0	0	0	1	0
C	0	0	0	1	0
D	0	0	0	0	1
E	0	0	0	0	0

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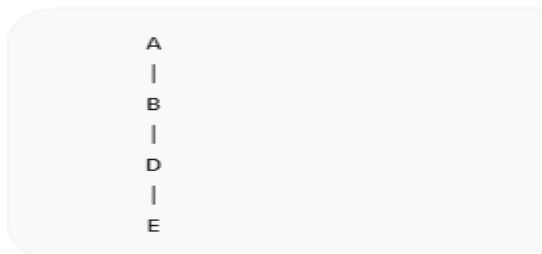
5. Deletion of a Node

Delete node C

Before Deletion



After Deletion



Remove C's row and column.

	A	B	D	E
A	0	1	0	0
B	0	0	1	0
D	0	0	0	1
E	0	0	0	0

```
def add_nodes(v):
    if v in graph:
        print(v, "already present")
    else:
        graph[v]=[]
def add_edges(v1,v2):
    if v1 not in graph:
        print(v1, "not present")
    elif v2 not in graph:
        print(v2, "not present")
    else:
        graph[v1].append(v2)
        graph[v2].append(v1)
def delete_edge(v1,v2):
    if v1 not in graph:
        print(v1, "not present")
    elif v2 not in graph:
        print(v2, "not present")
    else:
        if v2 in graph[v1]:
            graph[v1].remove(v2)
            graph[v2].remove(v1)
```

```
def delete_nose(v):
    if v not in graph:
        print(v, "not present in graph")
    else:
        graph.pop(v)
        for i in graph:
            list1=graph[i]
            if v in list1:
                list1.remove(v)
graph={}
add_nodes("A")
add_nodes("B")
add_nodes("C")
add_nodes("D")
add_edges("A", "B")
add_edges("B", "C")
add_edges("C", "A")
add_edges("A", "D")
add_edges("B", "D")
add_edges("C", "D")

print(graph)
```

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```
if v not in graph:
    print(v, "not present in graph")
else:
    graph.pop(v)
    for i in graph:
        list1=graph[i]
        if v in list1:
            list1.remove(v)

graph={}
add_nodes("A")
add_nodes("B")
add_nodes("C")
add_nodes("D")
add_edges("A", "B")
add_edges("B", "C")
add_edges("C", "A")
add_edges("A", "D")
add_edges("B", "D")
add_edges("C", "D")

print(graph)
```



... {'A': ['B', 'C', 'D'], 'B': ['A', 'C', 'D'], 'C': ['B', 'A', 'D'], 'D': ['A', 'B', 'C']}